SC400 Colorimeter Chlorine

0.01 to 2.00 mg/L
0.1 - 8.0 mg/L

Instruction Manual

Technical Data

Light source:	LED, Filter (λ = 528 nm)
Battery:	9 V battery (Life 600 tests without display light).
Auto-OFF:	Automatic switch off 5 minutes after last keypress
Ambient conditions:	5-40°C 30 - 90 % rel. humidity (non-condensing).
CE:	DIN EN 55 022, 61 000-4-2, 61 000-4-8, 50 082-2, 50 081-1, DIN V ENV 50 140, 50 204

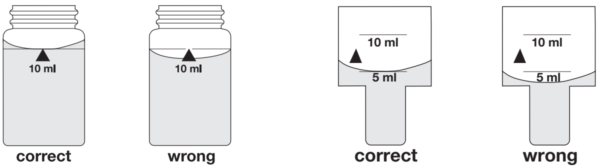
Method notes

Prior to measurement ensure that the sample is suitable for analysis (no major interferences) and does not require any preparation i.e. pH adjustment, filtration etc.

Reagents are designed for use in chemical analysis only and should be kept well out of the reach of children. Ensure proper disposal of reagent solutions.

Material Safety Data Sheets: available on request.

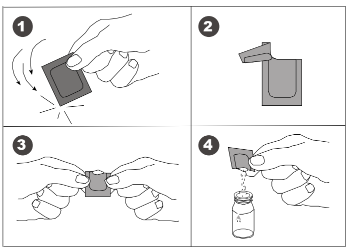
Correct filling of sample vial



Glass vial ø 24 mm

Plastic vial ⊔ 10 mm

Opening powder packs



Operation

On/Off

Switch the unit on using the ON/OFF key.

CLP1

The display shows the following:

Attention:
The unit is designed to measure chlorine in two different ranges.
Please select the required chlorine range as described below.
CLP1:10 mL sample **glass vial** ø 24 mm Range: 0.01 - 2.00 mg/L
CLP2: 5 mL sample **plastic vial** ⊔ 10 mm Range: 0.1 - 8.0 mg/L

Mode

Select the test required using the MODE key:
CLP1 → CLP2 → Abs → CLP1 → .... (Scroll)

METHOD

The display shows the following:

Fill a clean vial with the water sample up to the 5/10 mL mark (5 mL for 0-2 mg/L range, 10 mL for 0-8 mg/L range), cap and place the vial in the sample chamber with the Δ-mark on the vial aligned with the ∇–mark on the instrument.

Zero/Test

Press the ZERO/TEST key.

METHOD

The method symbol flashes for approx. 8 seconds.

0.0.0

The display shows the following:
After zero calibration is completed, remove the vial from the sample chamber.
Add the appropriate reagent; a color will develop in the sample.

Zero/Test

Cap and place the vial in the sample chamber with the Δ and ∇ marks aligned.

Press the ZERO/TEST key.

METHOD

The method symbol flashes for approx. 3 seconds.

RESULT

The result appears in the display.

The result is saved automatically.

Zero/Test

Repeating the analysis:

Press the ZERO/TEST key again.

Mode

New zero calibration:

Press the MODE key until the desired method symbol appears in the display again.

Measuring the absorbance

Abs

This Mode-function allows to measure the absorbance at the specified wavelength.
The result appears in the display in mAbs (1000 mAbs = 1 A (Absorbance unit)).

Display backlight

!

Press the "!" key to turn the display backlight on or off. The backlight is switched off automatically during the measurement.

Chlorine 0.01 - 2.00 mg/L with Powder Packs (CLP1)

with 24 mm glass vial

CLP1

0.0.0

(a) Free Chlorine

Perform zero calibration (**10 mL sample**, see "Operation").

Remove the vial from the sample chamber. Add the contents of one DPD Free Chlorine Powder Pack (10 mL sample). Cap and swirl the vial gently for 20 seconds. Replace the vial in the sample chamber immediately making sure the Δ and ∇ marks are aligned.

Zero / Test

CLP1

RESULT

Press the ZERO/TEST key.

The method symbol flashes for approx. 3 seconds.

The result is shown in the display in mg/L Free Chlorine.

Rinse the vial and cap thoroughly after each test.

(b) Total Chlorine

Perform zero calibration (**10 mL sample**, see "Operation").

Remove the vial from the sample chamber. Add the contents of one DPD Total Chlorine Powder Pack (10 mL sample). Cap and swirl the vial gently for 20 seconds. Replace the vial in the sample chamber making sure the Δ and ∇ marks are aligned.

Zero / Test

CLP1

RESULT

Wait for a color reaction time of 3 minutes.

Press the ZERO/TEST key.

The method symbol flashes for approx. 3 seconds.

The result is shown in the display in mg/L Total Chlorine.

Rinse the vial and cap thoroughly after each test.

(c) Combined Chlorine

Combined Chlorine = Total Chlorine - Free Chlorine

Measuring tolerance: 0-1 mg/L: ± 0.05 mg/L
\> 1-2 mg/L: ± 0.10 mg/L

Notes

1. Wipe liquid off the outer surface of the vial.
2. Do not use the same sample vial for free and Total Chlorine without thoroughly rinsing the vial between the two different tests.
3. Gentle swirling dissipates bubbles which may form in samples containing dissolved gases.

Chlorine 0.1 - 8.0 mg/L with Powder Packs (CLP2)

with 10 mm plastic vial

CLP2

0.0.0

(a) Free Chlorine

Fill a clean plastic vial with the water sample up to the 5 mL mark. Perform zero calibration (**5 mL sample, plastic vial**, see "Operation").

Remove the plastic vial from the sample chamber. Add the contents of **two** DPD Free Chlorine Powder Packs. Cap and swirl the plastic vial gently for 20 seconds. Replace the plastic vial in the sample chamber immediately making sure the Δ and ∇ marks are aligned.

Proceed immediately.

Zero/Test

Press the ZERO/TEST key.

CLP2

The method symbol flashes for approx. 3 seconds.

RESULT

The result is shown in the display in mg/L Free Chlorine.

Rinse the plastic vial and cap thoroughly after each test.

(b) Total Chlorine

Fill a clean plastic vial with the water sample up to the 5 mL mark. Perform zero calibration (**5 mL sample, plastic vial**, see "Operation").

Remove the plastic vial from the sample chamber. Add the contents of **two** DPD Total Chlorine Powder Packs. Cap and swirl the plastic vial gently for 20 seconds. Replace the plastic vial in the sample chamber making sure the Δ and ∇ marks are aligned.

Wait for a color reaction time of 3 minutes.

Zero/Test

Press the ZERO/TEST key.

CLP2

The method symbol flashes for approx. 3 seconds.

RESULT

The result is shown in the display in mg/L Total Chlorine.

Rinse the plastic vial and cap thoroughly after each test.

(c) Combined Chlorine

Combined Chlorine = Total Chlorine - Free Chlorine

Measuring tolerance:

• 2 - 3 mg/L: ± 0.2 mg/L • 3 - 4 mg/L: ± 0.3 mg/L • 4 - 8 mg/L: ± 0.4 mg/L

Notes

1. Wipe liquid off the outer surface of the vial.
2. Do not use the same sample vial for free and Total Chlorine without thoroughly rinsing the vial between the two different tests.
3. Gentle swirling dissipates bubbles which may form in samples containing dissolved gases.
4. If the chlorine concentration is less than 2 mg/L, use the 0.01 - 2.00 mg/L range (CLP1)

Menu selections

Mode

Press the MODE key and hold.

On/Off

Switch the unit on using the ON/OFF key.
Allow the 3 decimal points to be displayed before releasing the MODE key.

!

The “!” key allows for selection of the following menu points:

- ▲1 recall stored data
- ▲2 ▼3 setting the date and time
- ▼4 user calibration

The selected menu is indicated by an arrow in the display.

Mode

Confirm the selection with the MODE key.

Recall of stored data

The meter shows the most recent measurements taken in the following format (automatically proceeds every 3s until result is displayed):

Number	n xx (xx: 16...1)
Year	YYYY (i.e. 2007)
Date	mm.dd (monthmonth:dayday)
Time	hh:mm (hourhour:minuteminute)
Test	Method
Result	x,xx

Zero/Test

The ZERO/TEST key repeats the current data set.

Mode

The MODE key scrolls through all stored data sets.

!

Quit the menu by pressing “!” key.

Setting date and time (24-hour-format)

Mode

After confirming the selection with the MODE key the value to be edited will be shown for 2 sec.

SET

DATE

YYYY

(2 sec.)

The setting starts with the year (YYYY) followed by the actual value to be edited. Same applies for month (mm), day (dd), hour (hh) and minutes (mm). Set the minutes first in steps of 10, press the "!" key to continue setting of minutes in steps of 1.

Mode

Increase the value by pressing the MODE key.

Zero/Test

Decrease the value by pressing ZERO/TEST key.

!

Proceed to the next value to be edited by pressing “!” key.
After setting the minutes and pressing the “!” key the display will show “IS SET” and instrument returns into the measurement mode.

When the battery is taken of for more than 1 minute, the unit will automatically enter the date/time menu when switched on again.

User calibration

Note:

cAL	user calibration (Display in calibration mode)
CAL	factory calibration (Display in calibration mode)
Mode	Press MODE key and hold.
On/Off	Switch the unit on using the ON/OFF switch, Release the MODE key after approx. 1 second. Select the required menu points using the "!" key.
!	
CAL	After confirming the selection with the MODE key the instrument will show CAL/CLP1.
CLP1	Scroll through methods using the MODE key.
	Perform zero calibration (see "Operation").
Zero/Test	Press the ZERO/TEST key.
METHOD	The method symbol flashes for approx. 8 seconds.
0.0.0	The display shows the following in alternating mode:
CAL	Perform calibration with a standard which concentration is known (see "Operation").
Zero/Test	Press the ZERO/TEST key.
METHOD	The method symbol flashes for approx. 3 seconds.
RESULT	The result is shown in the display, alternating with CAL.
CAL	If the reading corresponds with the value of the calibration standard (within the specified tolerance), exit calibration mode by pressing the ON/OFF key.
	Changing the shown value:
Mode	Otherwise, pressing the MODE key once increases the displayed value by 1 digit.
Zero/Test	Pressing the ZERO/TEST key once decreases the displayed value by 1 digit.
CAL	Press the corresponding key until the reading equals the value of the calibration standard.
RESULT + x	
On/Off	By pressing the ON/OFF key, the new correction factor is calculated and stored in the user calibration software.
: :	Confirmation of calibration (3 seconds).

Recommended calibration value for CIP

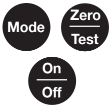
CLP1	between 0.5 and 1.5 mg/L
CLP2	between 3.0 and 5.0 mg/L

Factory Calibration Reset

Resetting the user calibration to the factory default will reset all methods and ranges.

A user calibrated method will be indicated by an arrow in the display.

To reset the calibration to the factory setting:



Press both the MODE and ZERO/TEST key and hold.

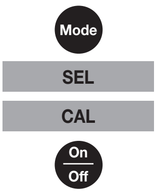
Switch the unit on using the ON/OFF key. Release the MODE and ZERO/TEST keys after approx. 1 second.

The following messages will appear in turn on the display:

The factory setting is active.
(SEL stands for Select)

or:

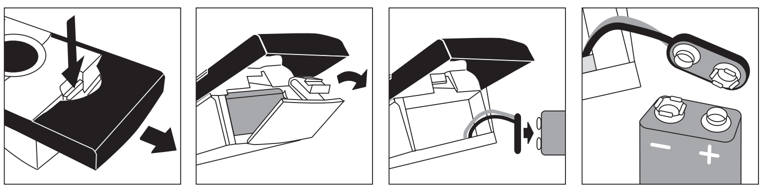
Calibration has been set by the user.
(If the user calibration is to be retained, switch the unit off using the ON/OFF key.)



Calibration is reset to the factory setting by pressing the MODE key.
The following messages will appear in turn on the display:

Switch the unit off using the ON/OFF key.

Changing the battery



Error codes

General error codes

EOOI	Light absorption too great. Reasons: dirty optics.
HI	Measuring range exceeded or excessive turbidity.
LO	Result below the lowest limit of the measuring range.
LO BAT	Replace 9 V battery, no further tests possible.
EOIO	Calibration factor "out of range"
E20	Oversupply of light reaches the detector

Individual error codes

E 70	CLP1:	Factory calibration incorrect / erase
E 71	CLP1:	User calibration incorrect / erase
E 72	CLP2:	Factory calibration incorrect / erase
E 73	CLP2:	User calibration incorrect / erase

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